Determinant of matrix times a constant.

Asked 8 years, 9 months ago Modified 5 years, 6 months ago Viewed 57k times

18

Prove that $\det(kA) = k^n \det(A)$ for and $(n \times n)$ matrix.

I have tried looking at this a couple of ways but can't figure out where to start. It's confusing to me since the equation for a determinant is such a weird summation.

linear-algebra matrices

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edited Apr 1, 2015 at 8:55
user99914

asked Apr 1, 2015 at 8:45
user139985
931 1 10 28

2 Can you explain how you have defined determinant? – Prahlad Vaidyanathan Apr 1, 2015 at 8:46

Also, do you know any other features of the determinant, e.g., if you have produce a matrix A' by replacing a row of a matrix A with k times that row, do you know how $\det A$ and $\det A'$ are related? – Travis Willse Apr 1, 2015 at 8:48

You can prove this directly, just by using the definition of the determinant. – peter.petrov Apr 1, 2015 at 8:52

4 Answers

Sorted by: Highest score (default)

24

A mnemonic using more advanced stuff, just to recover it from the few determinant rules I manage to remember is:

$$Det(kA) = Det(kI_n A) = Det(kI_n)Det(A) = Det\left(\begin{bmatrix} k & 0 & \dots & 0 \\ 0 & k & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & k \end{bmatrix}\right) Det(A)$$

$$= k^n Det(A)$$

Where I have used $Det(AB) = Det(A)Det(B)$ and a formula for the determinant of diagonal matrices.

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edited Jul 12, 2018 at 9:59

answered Jul 12, 2018 at 9:19
mihb
251 2 3

11

To elaborate on the above answer, the formula for the determinant is

$$\det(A) = \sum_{\sigma \in S_n} sign(\sigma)\Pi_{i=1}^n a_{i,\sigma_i}$$

so

$$\det(kA) = \sum_{\sigma \in S_n} sign(\sigma)\Pi_{i=1}^n k a_{i,\sigma_i}$$

$$= k^n \det(A)$$

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answered Apr 1, 2015 at 8:53
Ellya
11.7k 3 22 40

what if the matrix A is orthogonal? I'm confused because if A is orthogonal won't kA also be orthogonal and hence $det(kA) = +1$ or -1 ? – momo Sep 18, 2018 at 11:54

@momo No: if A is orthogonal then $A^T A = I$, so then $(kA)^T(kA) = k^2 I$. – Eric Kightley Jan 21, 2019 at 15:59

5

Note that the matrix kA has elements $[kA]_{ij} = kA_{ij}$, where A_{ij} are the elements of A. If we were to calculate the determinant expression formula, each term has the factor k appearing n times, where n is the dimension of the matrix. You can factor these out from the entire expression, and you're left with something proportional to $\det A$.

Example:

$$\begin{aligned}\det\left(k\begin{bmatrix}A_{11} & A_{12}\\ A_{21} & A_{22}\end{bmatrix}\right) =& \det\begin{bmatrix}kA_{11} & kA_{12}\\ kA_{21} & kA_{22}\end{bmatrix}\\ =& kA_{11}kA_{22} - kA_{21}kA_{12}\\ =& k^2(A_{11}A_{22} - A_{21}A_{12})\\ =& k^2 \det A\end{aligned}$$

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edited Apr 1, 2015 at 8:53

answered Apr 1, 2015 at 8:48
co9olguy
897 6 23

2

In the most simplest of case, say a matrix A which is a 2×2 matrix, multiplying it by a constant k gives the following general setup.

$$k\begin{pmatrix}a & b\\ c & d\end{pmatrix} = \begin{pmatrix}ka & kb\\ kc & kd\end{pmatrix}$$

The determinant is therefore (writing $B = kA$);

$$\begin{aligned}\det B &= (ka)(kd) - (kb)(kc)\\ &= k^2 ad - k^2 bc\\ &= k^2(ad - bc)\\ &= k^2 \det A\end{aligned}$$

Notice the exponent on k is of order 2 for the 2×2 case; a conjecture is that that is would be 3 for the 3×3 case. It is now a case of trying to expand that notion to the $n \times n$ case.

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edited Apr 1, 2015 at 11:39

answered Apr 1, 2015 at 8:59
Autolatry
2,996 1 15 16